📄 Fake News Detection System – Project Report

🧾 1. Title Page

Project Title: Fake News Detection System
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📌 2. Abstract

This project focuses on building a Fake News Detection System using Machine Learning techniques. The system takes news text as input and predicts whether it is REAL or FAKE. The model is trained using Natural Language Processing (NLP) techniques such as TF-IDF Vectorization and Logistic Regression. A web-based interface is also developed using Flask to allow users to interact with the model easily.

🎯 3. Objective

- To detect fake news using machine learning
- To preprocess textual data effectively
- To build a classification model
- To deploy the model using a web application

⚙️ 4. Technologies Used

- Python
- Pandas
- Scikit-learn
- Flask
- Pickle

🧠 5. Methodology

5.1 Data Collection

A dataset containing news articles labeled as REAL or FAKE was used.

5.2 Data Preprocessing

- Removed missing values
- Combined title and text
- Converted text to lowercase
- Removed punctuation

5.3 Feature Extraction

TF-IDF (Term Frequency - Inverse Document Frequency) is used to convert text into numerical form.

5.4 Model Training

- Algorithm: Logistic Regression
- Data split: 80% training, 20% testing

5.5 Model Evaluation

Accuracy score is used to evaluate model performance.

🏗 6. System Architecture

User Input → Preprocessing → TF-IDF → Model → Prediction (REAL/FAKE)

💻 7. Implementation

Modules:

- preprocess.py → Data cleaning
- train_model.py → Model training
- predict.py → Prediction via terminal

- app.py → Web interface

🌐 8. Web Application

A simple web app is created using Flask where users can input news text and get predictions instantly.

📊 9. Results

- The model achieved an accuracy of approximately **80–90%** (depends on dataset)
- Successfully predicts real and fake news

⚠️ 10. Limitations

- Small dataset reduces accuracy
- Cannot detect very complex fake news
- Depends heavily on training data quality

🚀 11. Future Scope

- Use larger datasets
- Apply deep learning (LSTM, BERT)
- Improve UI/UX
- Deploy as mobile/web app

📚 12. Conclusion

The Fake News Detection System demonstrates how machine learning can be used to solve real-world problems. It successfully classifies news articles and provides a foundation for more advanced systems.

📎 13. References

- Scikit-learn Documentation

- Pandas Documentation
- Flask Documentation